

UNDP DATA DRIVE HACKATHON 2025

DEVELOPMENT EFFECTIVE ANALYSIS TRACK

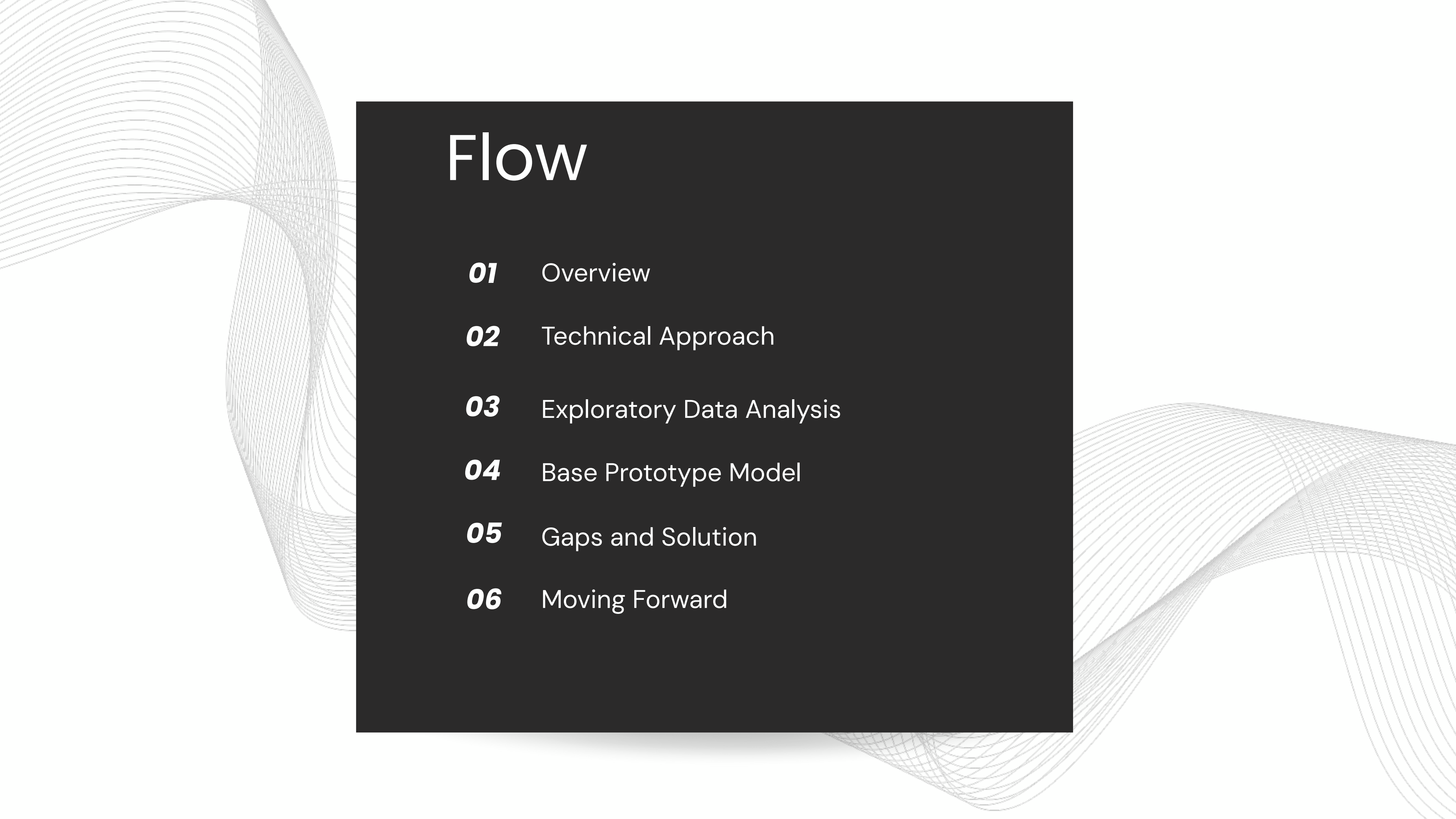
GENDER X CLIMATE MODEL

**PREDICTIVE ANALYTICS FOR
CROSS-SECTOR AID SYNERGIES**

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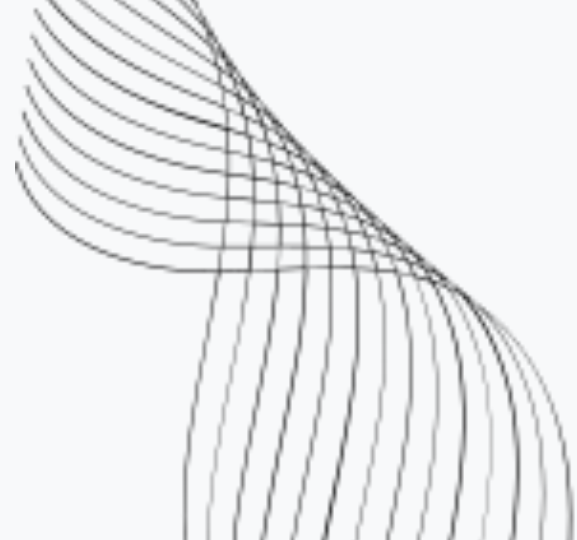
Hayat Hassanpour

Jostin Jerico Rosal



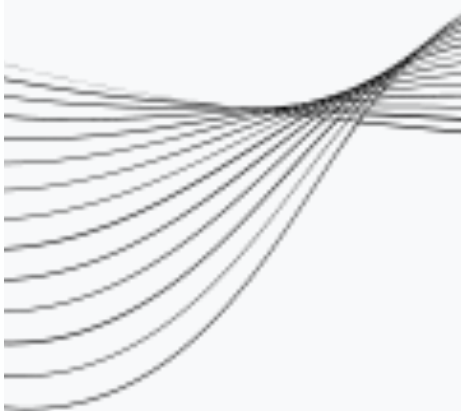
Flow

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Overview

Throughout the years, there have been continuous calls to strengthen both gender equality and climate action, yet these priorities are often treated separately when it comes to aid allocation.

- The 2015 Paris Agreement calls for gender-responsive climate action, but funding streams remain largely siloed.
 - The 2019 UNFCCC Gender Action Plan emphasizes the integration of gender considerations into climate policies, yet data-driven allocation models to operationalize this integration are still missing.
 - The UN Sustainable Development Goals explicitly link Gender Equality and Climate Action, urging innovative financing mechanisms to address their intersection.
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Despite these efforts, consistent and coordinated funding that leverages the synergies between gender and climate initiatives **remains insufficient**.



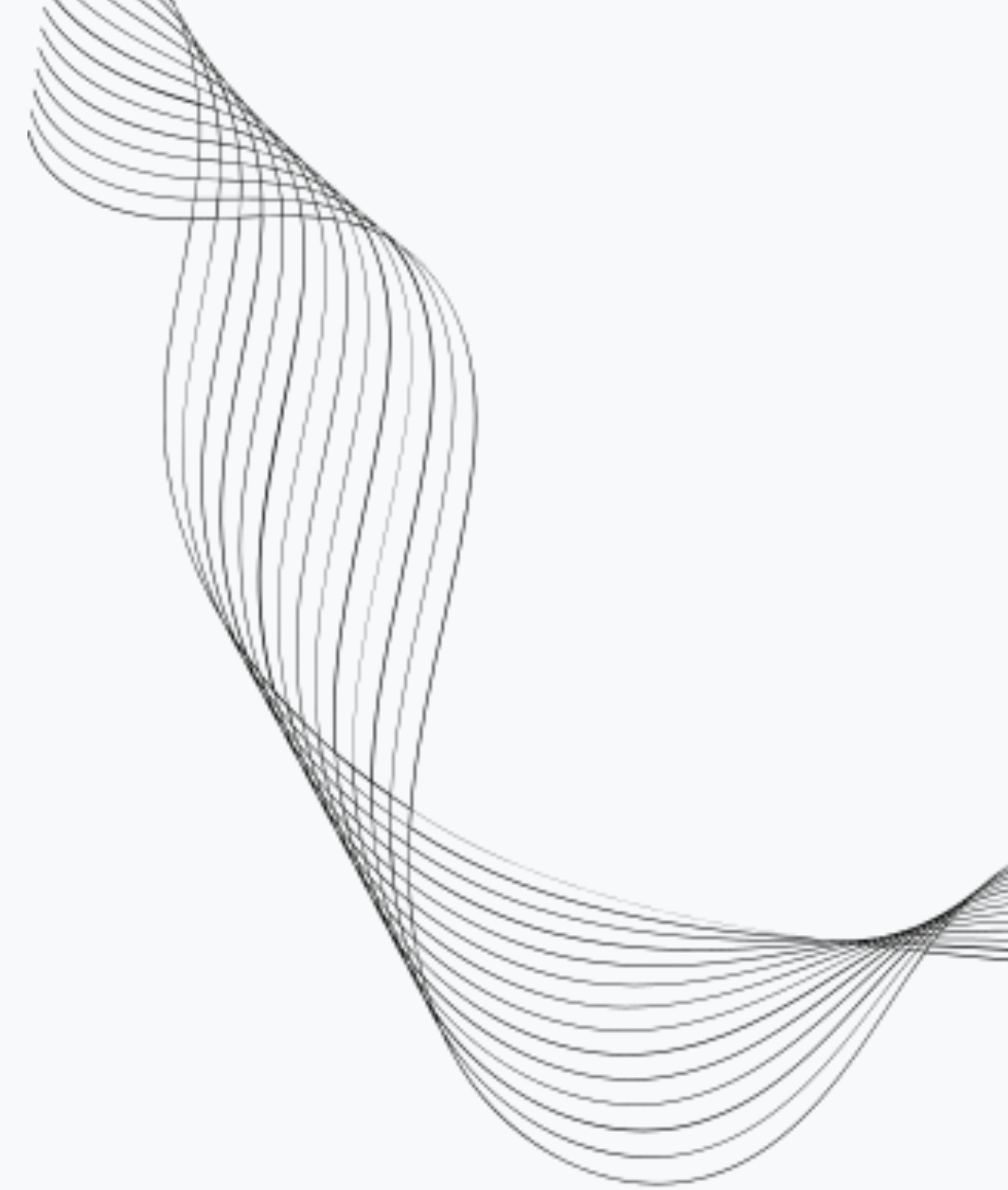
Technical Approach

Datasets

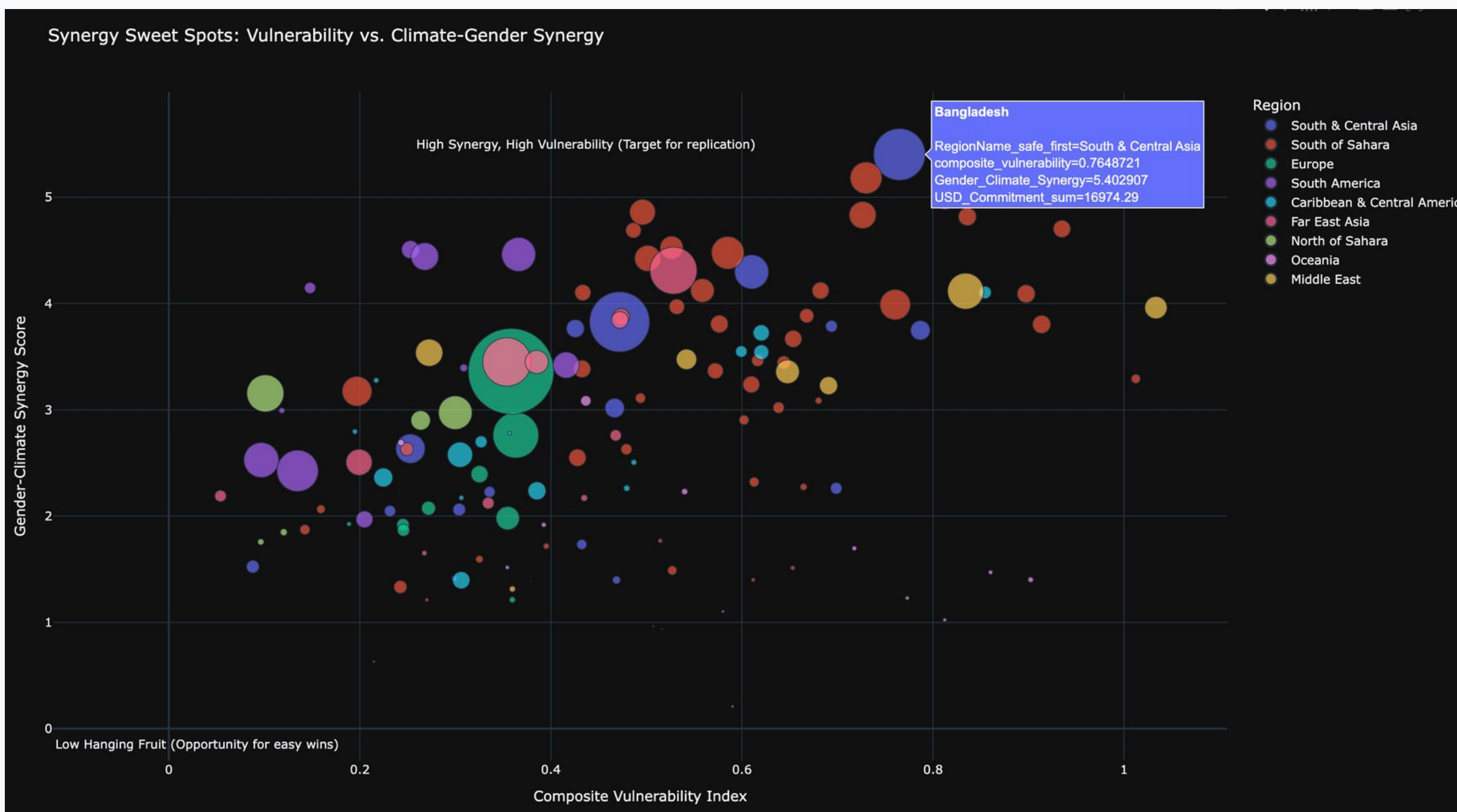
- **Primary Dataset:** OECD CRS Dataset
- **Secondary Datasets:**
 - UNDP HDR Dataset - Human Development Report Dataset
 - ND-GAIN Dataset (Notre Dame - Global Adaption Initiative Dataset)
 - AID Dataset
 - CCVI Dataset (Climate Conflict Vulnerability Index)

Tools used

- **Tools:** Jupyter, Python (Pandas, Statsmodels, Scikit-learn)
- **AI Usage:** ChatGPT/Claude for data cleaning and code reasoning



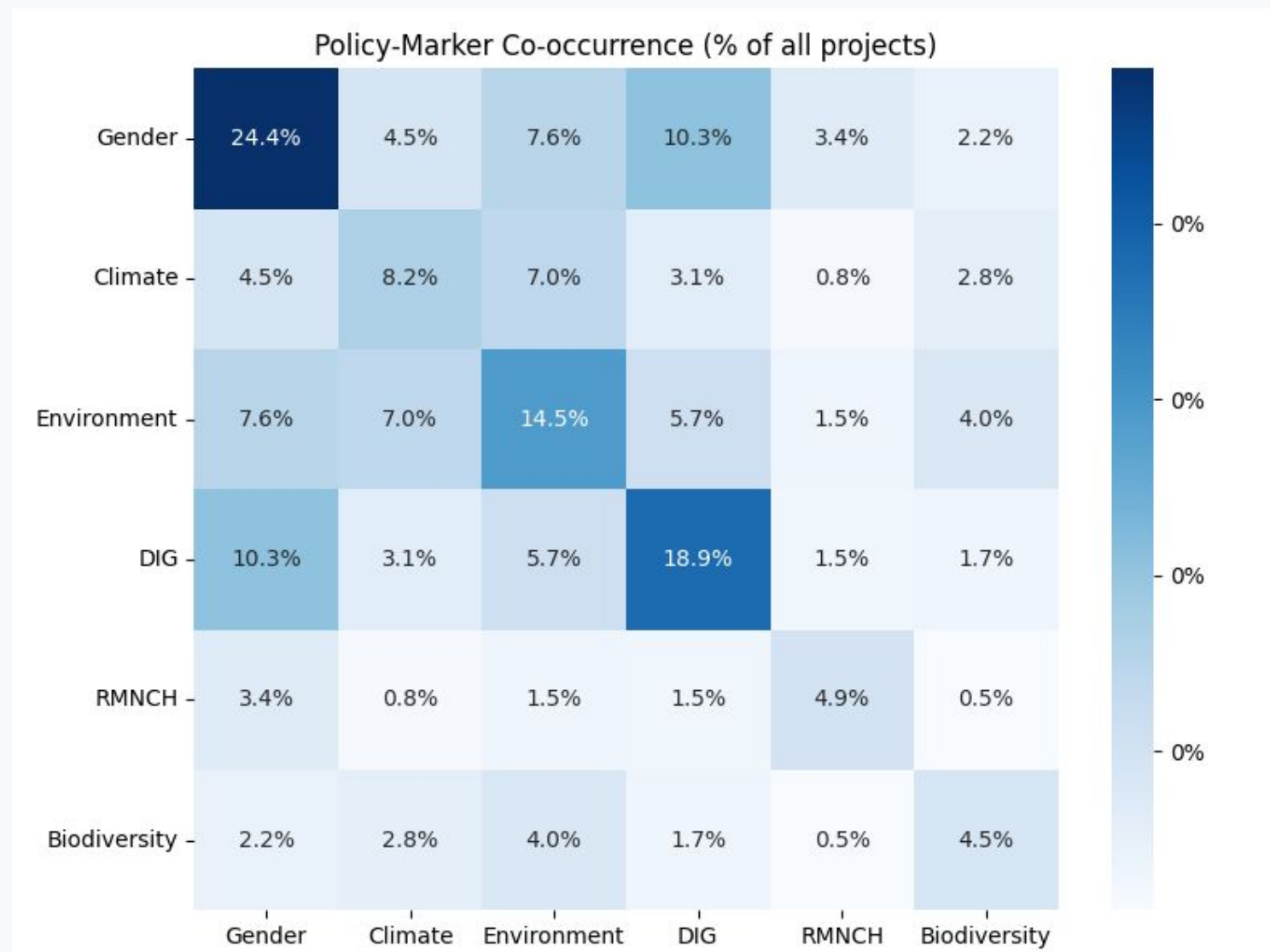
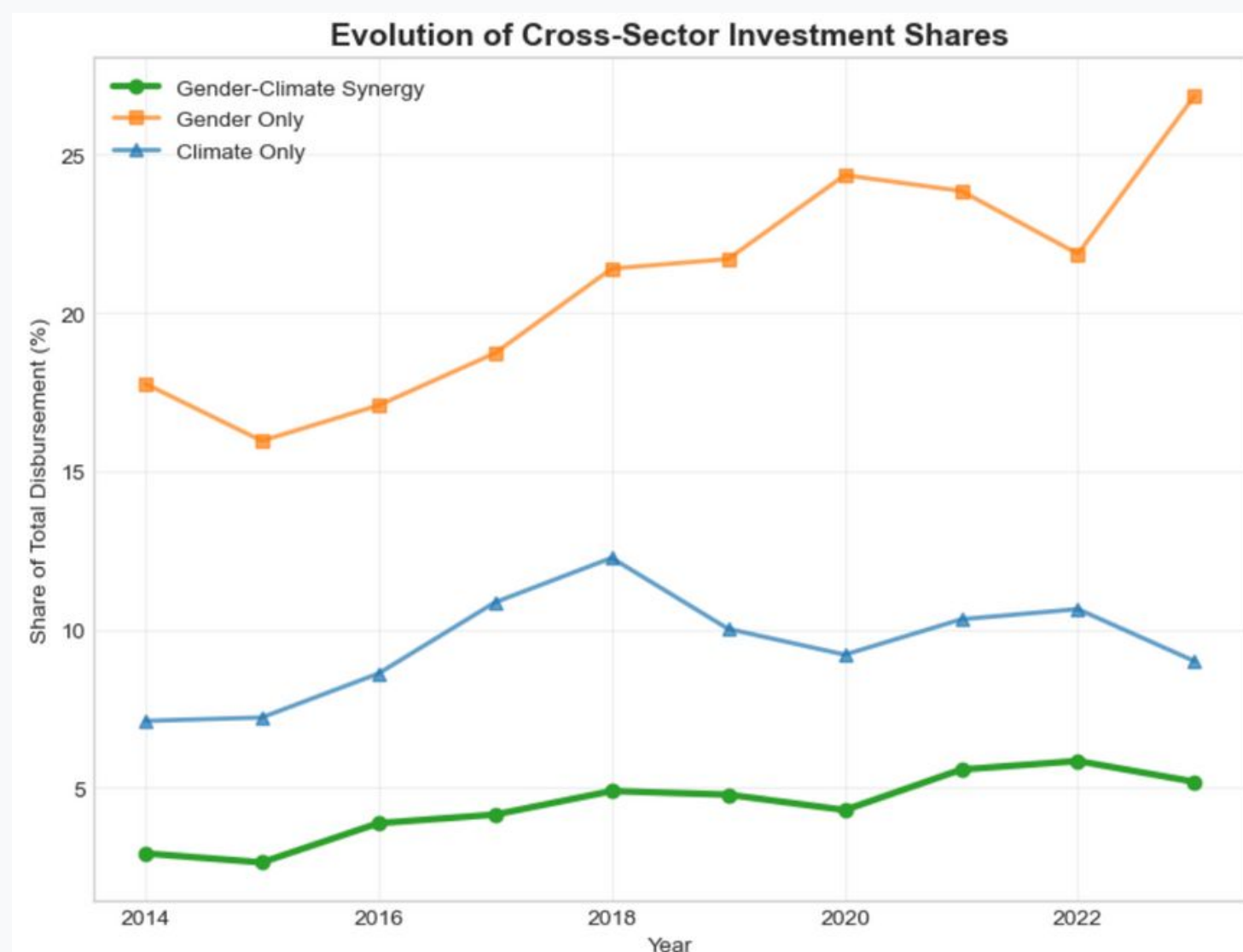
Climate-Gender Synergy vs Vulnerability Index



This map positions the countries that are high priority targets for crossover designs.

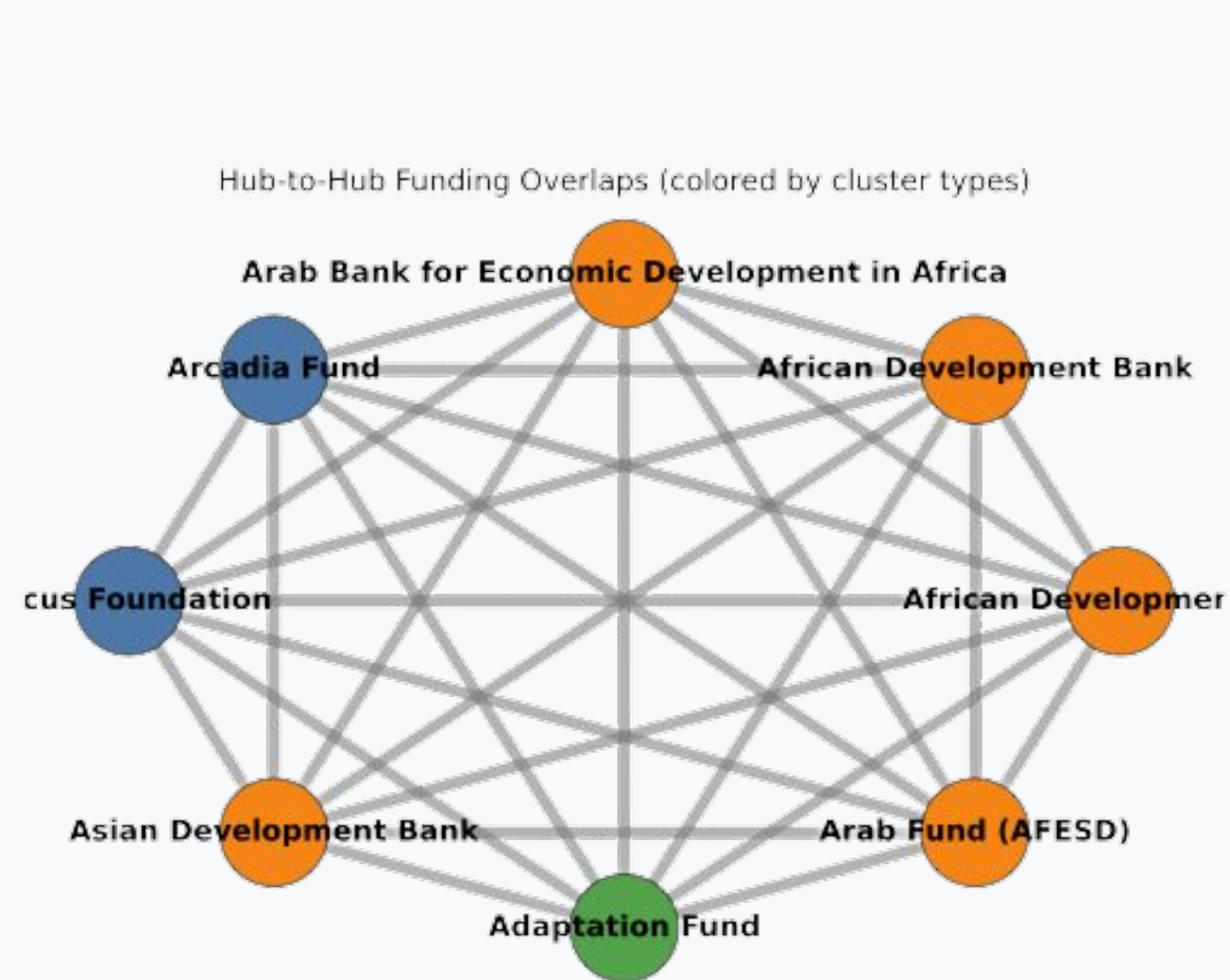
If we look Bangladesh, it is recognized as high vulnerable country but exhibits an usually high climate gender synergy score indicating hybrid projects would more likely have greater impact.

Cross Sector Analysis



Over the years, the disbursement towards climate related projects has been higher compared to gender related projects. there is still a **huge opportunity to leverage disbursements in cross sector projects**. from the insight analysis only **4.5%** of all projects carry gender-climate cross markers

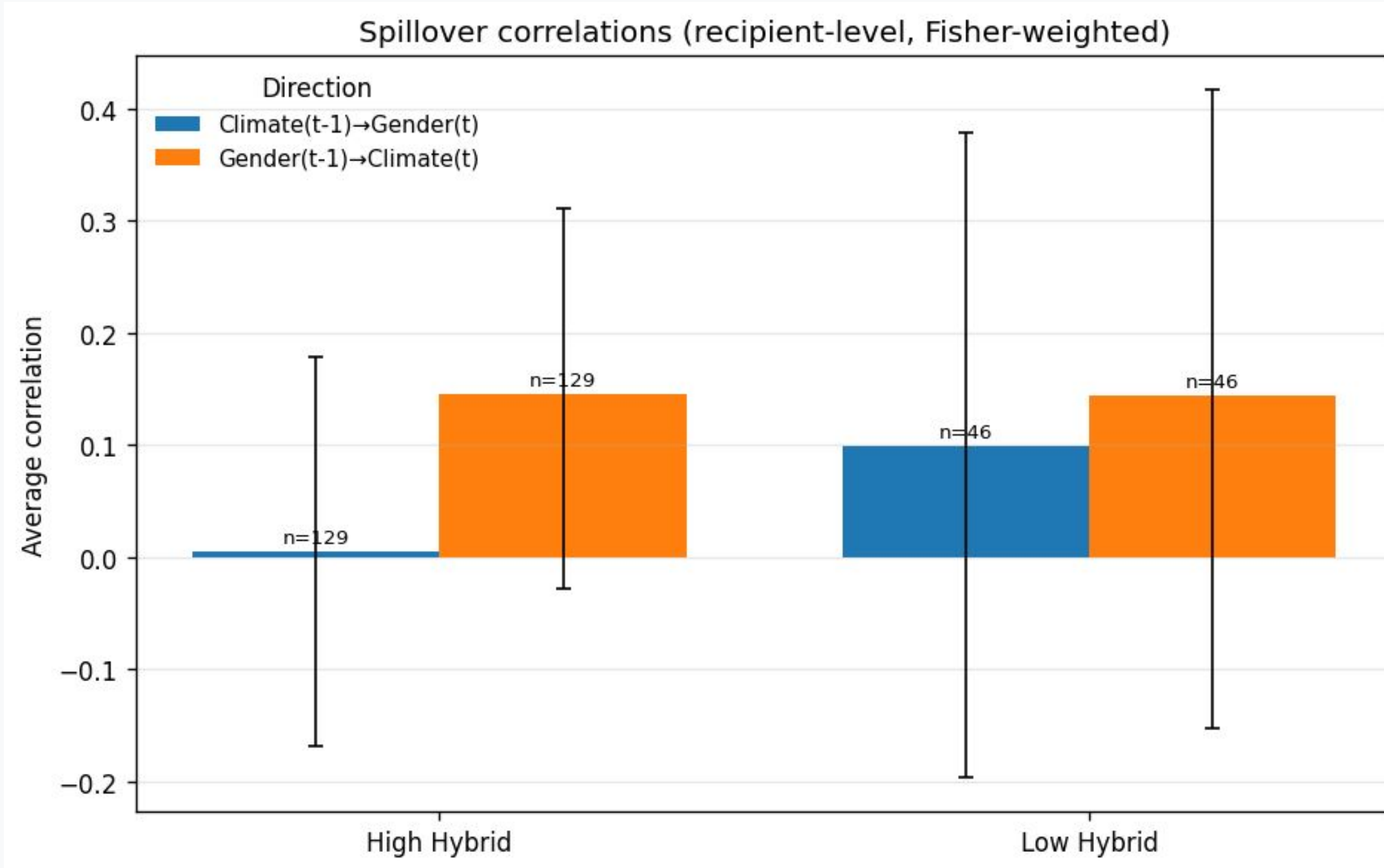
Hub-to-Hub Linkage Analysis



	Values
Cross - Cluster Overlap	60.7
Assortativity Index	-0.143
HHI (Pair Concentration)	0.036

Amongst the highest-linked donor hubs, majority are under-tagged with only 2 in climate leaning clusters and only 1 hub leads to co-benefit funding. Even though total overlaps between different hubs composes about 60%, the pair concentration is still low (~0.036) , indicating for viable corridors for collaboration.

Spillover Correlations

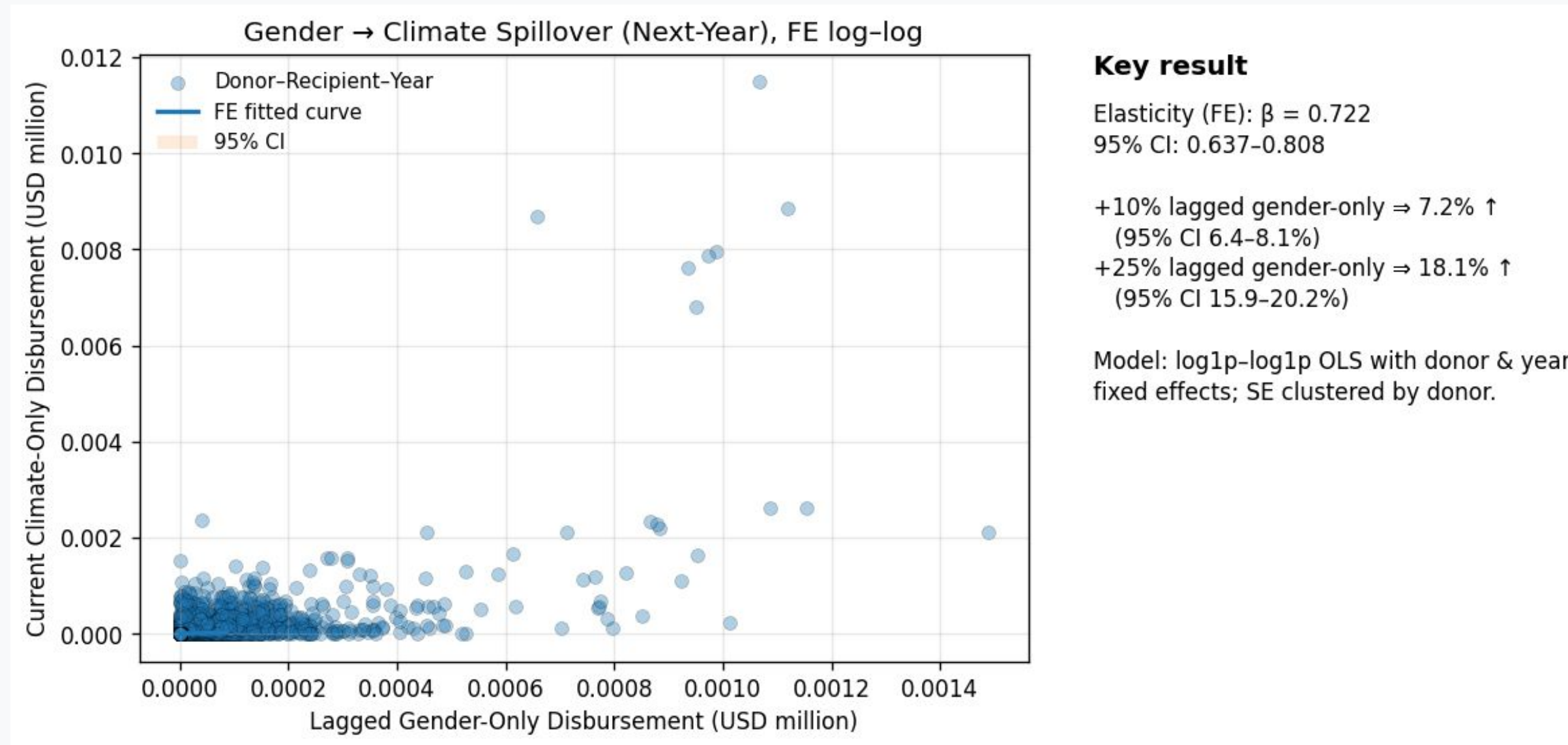


At the recipient level, the gender funding in the previous year turns be a better predictor of the consequent years' climate funding.

However, the the correlation bar's confidence intervals include zero due to limited sample and heterogeneity.

Thus, leading a fixed-effects model to check for significance.

Base Prototype – Regression Analysis



Controlling for who the donor is and the year, the analysis reveals a large and precise spill-over effect. A 10% increase in **Gender Only** Disbursement is associated with a ~7.2% increase of **Climate Only** Disbursement in the consequent year.



Gap

From the Analysis, Gender Focused and Climate Focused often cascade into one another, however, we currently lack a forward looking model that can maps donor - recipient relations before funds are committed.





Solution

We aim to implement a predictive model that makes use of historical funding, project share dynamics, important economic control factors to predict gender-climate funding relations.

Next Steps

- Developing an Advanced Machine Learning Model that will address the critical gap in cross sector funding.



THANK YOU

